

## TRD Pipeline:

- Filter patients, slice them by anchor date
- Vectorize Patients (2 methods)
- Split into train/test
- Use ML/KNN to predict TRD

## Counterfactual Pipeline:

- Filter patients, slice them by anchor date
- Vectorize patients (feature vector only)
- Split into train/test
- Have several treatment registries
  - e.g. Bupropion vs. SSRI
  - SNRI vs. SSRI
  - etc.

- For each treatment pair, select from train/test only relevant patients w/ one of two meds @ anchor date
  - Train model\_med1 to predict TRD on train set patients w/ med1, evaluate on test set patients w/ med1, USE on test set patients w/ med2
  - Vice versa

Each test patient:

risk if SNRI .55 (observed)  
risk if SSRI .4 (ungradeable)  
effect .15 (suggest SNRI better)

→ Average over all patients w/ confidence interval

You must defend this value

"What if medication A only looks better because A patients were healthier to begin with?"

Investigate:

★ Are features similar between the two arm populations? (Other measures?)

→ SMD = (mean in arm A - mean in arm B) / (pooled stdev)

★ Overlap/positivity e.g. if all 60+ yr old men only got one type of med)

→ Does every covariate profile have nonzero probability

→  $e(x) = P(\text{patient gets arm A} \mid \text{covariates } x)$

→ If  $e(x)$  is below or above a certain interval  $\leq [0, 1]$ , that patient does not contribute to ATE

→  $e(x)$  comes from a trained model

★ E-value (unmeasured confounding)

→ "What is the minimum strength of association that U would need w/ both treatment and outcome to reduce observed effect to nothing?"

→  $RR = \frac{P(\text{TRD} \mid \text{arm A})}{P(\text{TRD} \mid \text{arm B})}$

(if  $RR < 1$ , take  $RR_{\text{new}} = 1/RR$ )

E-value =  $RR + \sqrt{RR(RR-1)}$

U must be this strongly related to both treatment and outcome

| group       | N   | got arm A | reached TRD |
|-------------|-----|-----------|-------------|
| high # enc. | 100 | 70        | 40          |
| low #       | 100 | 40        | 20          |

$P(A \mid \text{high}) = 0.7$   
 $P(A \mid \text{low}) = 0.4$   
 $RR_1 = \frac{.7}{.4} = 1.75$

$P(\text{TRD} \mid \text{high}) = 0.4$   
 $P(\text{TRD} \mid \text{low}) = 0.2$   
 $RR_2 = \frac{.4}{.2} = 2$

★ Negative Control outcome

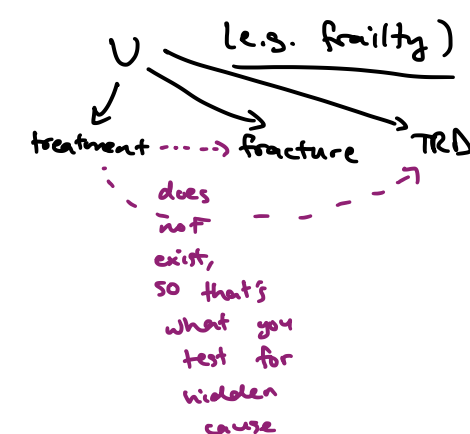
→ Pick outcome med choice can't plausibly cause

→ If method returns non-zero effect, indicates confounding w/ hidden cause affecting both outcome and med choice

e.g. TRD effect = 0.05

Bone spur effect = 0.04

.0001



num-encounters carries

(1.75, 2)

Feasible for supposed U to beat

→ Compute over all features

Worried:

∃ feature s.t. both measured arrows  $\geq$  E-value

Not worried:

∀ features, at least one of arrows  $<$  E-value

so feasibility an unknown U could as well